

# 1. Swarm UAV Competition | TEKNOFEST

## Competition Objective

- Develop **software and algorithms** for multiple UAVs operating as a coordinated **swarm**.
- Enable the UAV swarm to **complete designated tasks autonomously/cooperatively**.
- Demonstrate the developed algorithms using UAVs under **real-world conditions**.
- Encourage development and understanding of **Swarm UAV technology**.
- Explore potential **civilian and military applications**.

## Who Can Participate?

- University **students and graduates** can participate.
- **Team size:** Minimum **3**, maximum **10** members.

## Competition Evaluation Stages

The competition consists of **five main stages**:

1. **Preliminary Design Report**
2. **Preliminary Evaluation Presentation**
3. **Critical Design Report**
4. **Flight Evidence and Software Evaluation Video**
5. **Final Competition**

## IMP DATES IN 2026 , 2027 NOT ANNOUNCED YET

| Explanation                                                          | Date                     |
|----------------------------------------------------------------------|--------------------------|
| Competition Application Deadline                                     | 15.03.2026               |
| (PDR) Final Submission Date                                          | 27.03.2026 17:00 (GMT+3) |
| Announcement of PDR Results and Teams<br>Advancing to the Next Round | 01-04.04.2026            |

|                                                                            |                                |
|----------------------------------------------------------------------------|--------------------------------|
| Question and Answer Meeting                                                | 10.04.2026                     |
| Preliminary Assessment Presentation (Simulation)                           | 20-21.04.2026                  |
| Preliminary Assessment Presentation<br>Announcement of Elimination Results | 27-30.04.2026                  |
| Critical Design Report (CDR) Final Submission Date                         | 19.06.2026 / 17.00 (GMT+3)     |
| Announcement of CDR Results                                                | 06 - 10.07.2026                |
| Flight Evidence and Software Architecture Video Final Submission Date      | 02.08.2026                     |
| Announcement Date for Video Results—Finalists                              | 07.08.2026                     |
| Competition Date - Location                                                | 07 - 11.09.2026 - GAZIANTEP    |
| TEKNOFEST Şanlıurfa                                                        | September 30 - October 4, 2026 |

## Mission Flow

### 1. Autonomous Swarm Takeoff

- Minimum **3 UAVs**.
- All UAVs take off simultaneously.
- Only **one start command** is allowed.
- UAVs maintain the initial formation while climbing.

### 2. Navigate to QR Points

- Swarm autonomously travels between QR locations.
- Formation must be maintained during navigation.
- The formation rotates toward the direction of the next QR.
- UAV headings must adjust toward the travel direction.

### 3. QR Detection & Decoding

- At least **one UAV must have a camera**.

- At least one UAV detects and decodes the QR.
- The swarm selects the instructions corresponding to its **Team ID**.
- QR information determines the tasks and **next QR point**.

#### 4. **Formation Change**

- Swarm must dynamically switch between requested formations.
- Required spacing between UAVs must be maintained.

#### 5. **Pitch / Roll Maneuvers**

- Entire formation may be commanded to tilt around pitch or roll axes.
- Formation center remains fixed.
- After the maneuver, the swarm maintains the new angle until another formation/maneuver command changes it.

#### 6. **Altitude Change**

- Entire swarm moves to the requested altitude.
- Formation must remain intact.

#### 7. **Remove / Rejoin a UAV**

- QR specifies:
  - UAV ID to leave the swarm.
  - Red or blue landing zone.
- Selected UAV separates and lands in the specified zone.
- It disarms and waits.
- It must then re-arm and rejoin the swarm by the next QR task at the latest.
- Remaining UAVs must preserve their current formation.

#### 8. **Colored Zone Detection**

- Swarm must detect **red and blue landing zones while flying the route**.
- Their positions must be recorded for later use.
- No additional scanning is allowed specifically to find them.

#### 9. **Return Home**

- **Next QR = 0** means the final QR.
- Complete the QR task.
- Return autonomously to home.
- Land safely while maintaining formation.
- Disarm all UAVs.



## 2. TEKNOFEST Drone Championship | TEKNOFEST

No cz individual participation only no teams

### World Drone Cup | TEKNOFEST

this one is also individual

### international Unmanned Aerial Vehicle Competition | TEKNOFEST

#### Objective:

Develop students' UAV knowledge and technical skills, encourage innovative UAV projects, and bring together national and international students.

#### Participation:

Team-based competition for national and international associate, undergraduate, graduate, and eligible high-school students.

#### Categories:

- Fixed Wing
- Rotary Wing
- Free Mission

#### Evaluation:

- Project Presentation Report
- Preparation and Mission Video

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| Explanation                                         | Date                             |
|-----------------------------------------------------|----------------------------------|
| Competition Deadline                                | March 15, 2026                   |
| Project Presentation Report Submission              | March 9–16, 2026 - 17.00 (GMT+3) |
| Announcement of Project Presentation Report Results | April 13–17, 2026                |

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Submission of Commitment and Consent  
Forms by Teams Successful in the Project  
Presentation Report

April 21 – May 8, 2026

Deadline for Withdrawing from Competitions  
and Member Changes

June 5, 2026

Submission of Preparation and Mission  
Videos

July 13–17, 2026 - 17.00 (GMT+3)

Announcement of Preparation and Task  
Video Results

August 10–14, 2026

Competition Week and Awards Ceremony

September 8–13, 2026 / Malatya Tulga Havaalanı

TEKNOFEST

TEKNOFEST Şanlıurfa September 30 – October 4, 2026

| Fixed Wing                      | Rotary Wing               | Free Mission                            |                                                            |
|---------------------------------|---------------------------|-----------------------------------------|------------------------------------------------------------|
| <b>Aircraft type</b>            | Fixed-wing only           | Multirotor/helicopter (quad, hex, etc.) | Any type — fixed wing, VTOL, flapping wing, hybrid, rotary |
| <b>Max weight</b>               | 4 kg                      | 4 kg                                    | 25 kg                                                      |
| <b>Storage box shape</b>        | Cylindrical               | Rectangular/square prism                | None specified                                             |
| <b>Mission defined by</b>       | TÜBİTAK (fixed)           | TÜBİTAK (fixed)                         | The team itself                                            |
| <b># of missions</b>            | 2 (Mission 1 + Mission 2) | 2 mandatory + 1 optional bonus mission  | 1 self-designed mission, up to 3 flight attempts           |
| <b>VTOL allowed?</b>            | ✗ No                      | ✗ No                                    | ✓ Only here                                                |
| <b>Target detection (Int'l)</b> | Image processing required | Image processing required               | Team's choice                                              |

|                                       |                                             |                                                          |               |
|---------------------------------------|---------------------------------------------|----------------------------------------------------------|---------------|
| <b>Target detection (High School)</b> | Target location given (no image processing) | Image processing required (unlike Fixed Wing HS version) | Team's choice |
| <b>Max flight time (M2)</b>           | 10 min                                      | 10 min                                                   | 15 min total  |

## Competition Categories

1. **Fixed Wing** – autonomous fixed-wing aircraft
  2. **Rotary Wing** – multirotor/helicopter-type aircraft
  3. **Free Mission** – any UAV type (fixed wing, VTOL, flapping wing, hybrid, rotary), team defines its own mission
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## MISSIONS

### Fixed Wing Category (Int'l & Inter-High-School)

**Mission 1** – unloaded flight test:

- Take off, fly a figure-eight ( $\infty$ ) maneuver around two poles (2 laps) staying outside the poles
- Scored on: storage-box volume, install/prep time, unloaded weight, lap time, autonomy level
- Fail this → can't attempt Mission 2

**Mission 2** – payload drop:

- Same course, but now carrying 2 payloads (different weights, declared in advance)
- **International:** must *detect target locations via image processing* (targets' positions unknown) — red payload → 4×4 m blue target, blue payload → 2×2 m red target
- **Inter-High-School:** target locations are *given in advance* (no image processing needed) — blue payload → 10 m-diameter target, red payload → 2 m-diameter target
- Target-distance measured from where payload finally rests, up to 20 m from target center

### Rotary Wing Category (Int'l & Inter-High-School)

**Mission 1** – same figure-eight unloaded flight test as Fixed Wing, but with a rectangular/square storage box.

**Mission 2** – payload drop, both competitions use image processing to find target locations:

- **International:** red payload → blue hexagon (2 m sides), blue payload → red triangle (1 m sides); measured up to 10 m from center

- **Inter-High-School:** red payload → 2×2 m blue target, blue payload → 1×1 m red target

**Mission 3 (International only, optional)** – after dropping both payloads, return to the first target, *pick up* that payload, and drop it on the *other* target — earns bonus points, not mandatory.

### Free Mission Category

- No fixed mission — teams design their own scenario, described in the Project Presentation Report and Preparation/Mission Video
  - Same scenario must be reproduced exactly in the competition area
  - Max flight time: 15 minutes; 5-minute briefing to judges beforehand
  - Autonomous execution scores higher than manual
  - AAC can reject or require modification of a proposed mission on safety/area/rule grounds (e.g., paint gun instead of a real weapon mechanism, limits on high-speed aerobatics)
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## KEY RESTRICTIONS

### Team/eligibility

- Teams of 5–10 members; one Team Responsible required
- High-school students (Inter-High-School comp) or university/grad students (International); "mixed teams" allowed with <50% high schoolers
- One person can only be on one team, in one of the two competitions
- No first-degree kinship/conflict of interest between members and advisor

### Aircraft build & weight

- Fixed Wing / Rotary Wing: max take-off weight **4 kg**; design/build must be done by the team (no ready-made or replica UAVs)
- Free Mission: max take-off weight **25 kg**; doesn't need to be original design, but parts must be declared
- VTOL UAVs are **barred** from Fixed Wing/Rotary Wing — Free Mission only

### Safety-critical hardware

- Mandatory radio fail-safe (auto-triggers within 5 sec of signal loss)
- Mandatory emergency power cut-off switch (cuts power within 2 sec)
- End-to-end encrypted radio control and telemetry required
- Arm/disarm only via transmitter stick movements

### Flight rules

- Max altitude: 120 m AGL
- Wind threshold for suspension: 25 km/h

- Only 1 camera allowed for image processing — a second observation camera is **strictly prohibited**
- No jamming, spoofing, or interference with other teams' signals (instant disqualification)
- Must use same motor/propeller across Mission 1 and Mission 2

### Payload rules

- Payloads must be identical except color; only the specified paint color allowed (no QR codes, labels, stripes)
- No electronics inside payloads
- No parachute-type systems unless same color as payload
- Payloads must not break apart on impact (or scoring is voided)

### Prohibited materials/content

- **No biological, chemical, or hazardous substances** on any flight
- **No explosions or anything risking human/environmental safety** — paint or bead-gun systems are suggested substitutes for "weapon-style" Free Mission concepts



### 3. FPV Drone Tracking Competition | TEKNOFEST

#### Competition Objective:

The primary aim is to evaluate the image-based perception, decision-making, and dynamic target tracking capabilities of developed autonomous aerial vehicles within a realistic mission scenario. The competition focuses on successfully tracking a moving target using image processing and autonomous control algorithms, particularly **without the use of GNSS (Global Navigation Satellite System)**. Teams are expected to develop systems that can autonomously track a moving target drone using vision alone, building capabilities in advanced target tracking, frame centering, lost-target reacquisition, and dynamic flight performance.

#### Who Can Participate:

- University students (associate degree, bachelor's, master's, doctoral) studying in Türkiye or abroad
- Graduates
- Team size: **4–10 members**, including the advisor

#### MISSION

**Core task:** Autonomously lock onto and track a moving target drone through a set of **11 predetermined maneuvers**, without using GNSS.

#### How it runs:

1. Team **manually** takes off and flies into the designated mission area (target drone starts stationary)
2. Once locked on, pilot switches **their own drone to autonomous mode**
3. Target drone then begins moving through a common, pre-set maneuver sequence
4. Competing drone must **autonomously track it** through all 11 maneuvers
5. Live camera feed from the tracking drone is shown to the referee throughout
6. At the end, pilot exits autonomous mode and returns the drone to the starting point

#### Timing:

- Team must reach competition area within **5 minutes** of being called
- **15 minutes** prep time
- **10 minutes max** task duration (timer starts when target drone takes off)
- If time runs out mid-task, score = however many maneuver stations were completed

**Attempts:** One attempt per day, 4 competition days total = **4 attempts**. Requesting a restart **resets all points** earned that attempt.

**Scoring (Task = 75% of total, Reports/Video = 25%)**

| Task Step                    | Points |
|------------------------------|--------|
| Live image transmission      | 10     |
| Autonomous flight capability | 10     |
| Maneuver Station 1           | 10     |
| Maneuver Station 2           | 10     |
| Maneuver Station 3           | 15     |
| Maneuver Station 4           | 20     |

|                          | (Fixed Wing)                                        | (Rotary Wing)                   | (Free Mission)             | FPV Drone Tracking                               | Swarm UAV                            | Drone Championship                   | World Drone Cup                      |
|--------------------------|-----------------------------------------------------|---------------------------------|----------------------------|--------------------------------------------------|--------------------------------------|--------------------------------------|--------------------------------------|
| <b>Objective</b>         | Autonomous flight + accurate payload drop           | Same + optional pickup/relocate | Team-designed mission demo | Vision-based autonomous target tracking, no GNSS | Multi-UAV coordinated swarm behavior | Not detailed (racing-style, assumed) | Not detailed (racing-style, assumed) |
| <b>Participation</b>     | Team                                                | Team                            | Team                       | Team                                             | Team                                 | <b>Individual only</b>               | <b>Individual only</b>               |
| <b>Who can join</b>      | High school + university/grad (mixed teams allowed) | Same                            | Same                       | University students + graduates                  | University students + graduates      | Not specified                        | Not specified                        |
| <b>Team size</b>         | 5–10                                                | 5–10                            | 5–10                       | 4–10                                             | 3–10                                 | N/A                                  | N/A                                  |
| <b>Max weight</b>        | 4 kg                                                | 4 kg                            | 25 kg                      | UAV-1 class (4–25 kg)                            | Not specified per UAV                | Unknown                              | Unknown                              |
| <b>VTOL allowed?</b>     | ✗                                                   | ✗                               | ✓ only here                | Not applicable (fixed drone type)                | Not specified                        | Unknown                              | Unknown                              |
| <b>GNSS use</b>          | Allowed/expected                                    | Allowed/expected                | Allowed                    | <b>Explicitly banned</b>                         | Required for swarm nav               | Unknown                              | Unknown                              |
| <b># missions/stages</b> | 2 missions                                          | 2 + 1 bonus                     | 1 self-designed            | 1 task (11 maneuvers)                            | 8-step mission flow (takeoff →       | Unknown                              | Unknown                              |

|                               |                                                |                            |                                  |                                                          |                                              |                         |                         |
|-------------------------------|------------------------------------------------|----------------------------|----------------------------------|----------------------------------------------------------|----------------------------------------------|-------------------------|-------------------------|
|                               |                                                |                            |                                  |                                                          | return home)                                 |                         |                         |
| <b>Core skill tested</b>      | Precision drop + autonomy                      | Precision drop + retrieval | Creativity + engineering breadth | Real-time vision tracking w/o GPS                        | Multi-agent coordination, formation control  | Piloting speed/skill    | Piloting speed/skill    |
| <b>Evaluation stages</b>      | Report → Video → Live flight                   | Same                       | Same                             | Pre-Design → Critical Design → Video → Live flight       | PDR → Simulation → CDR → Video → Live flight | Unknown                 | Unknown                 |
| <b>Competition venue/date</b> | Malatya (Sept 8–13) + Şanlıurfa TEKNOFEST      | Same                       | Same                             | Şanlıurfa, Aug–Sept                                      | <b>Gaziantep</b> (Sept 7–11) + Şanlıurfa     | Unknown                 | Unknown                 |
| <b>1st place prize</b>        | 250,000 TRY                                    | 250,000 TRY                | 250,000 TRY                      | 300,000 TRY                                              | Not stated yet                               | Unknown                 | Unknown                 |
| <b>Technical difficulty</b>   | High (custom airframe, no off-the-shelf parts) | High                       | Medium–High (open design)        | High (vision-only tracking is a hard CV/control problem) | Very high (multi-agent systems + comms)      | Medium (piloting skill) | Medium (piloting skill) |